Pip: A Lightweight AI Assistant Window with Safe Local Tool Access

Category: iTECH — Software / Programming
Competition: iWISE Regional Final 2026, Yogyakarta
Participants: Haidar Ali Fawwaz Nasirodin, Faiq Abiyyu Asyam, Avandreda Giano Bestafa

Abstract

Students use AI chat daily, yet giving it their own documents is still manual. They copy text out of PDFs, Word files and slides, and the broken formatting burns free-tier limits. The AI cannot see their files, so every question needs pasted context. Autonomous agents would remove that friction, but they require paid API keys or capable hardware, which many students do not have.

Pip is a small always-on-top desktop window, built with Tauri and TypeScript, that opens the AI chat service a user already has beside a set of local tools. A local pipeline built on Microsoft's MarkItDown converts PDF, DOCX, PPTX and XLSX files, lets the user preview and edit the Markdown, and shows a token count for the prompt it assembles. A local Model Context Protocol server gives the AI controlled access to the user's files, guarded by a folder allowlist with path canonicalization, native confirmation prompts before any write, authenticated connections, output size limits and an append-only audit log, while the chat webview is deliberately granted no application permissions. A Pomodoro timer and quick notes sit alongside them.

Pip needs no API key and no local model. The full path from document to tool call was verified on a free Claude account through a public tunnel, and adversarial testing closed path traversal and unauthenticated access while exposing an approval bypass that a green test suite had missed. A separate measurement showed that conversion alone makes a prompt 12% larger, and that selecting two pages instead of six cuts it by 63%.

Keywords: AI assistant, Model Context Protocol, local tool access, permission boundary, Tauri